

Lancaster University College
at Beijing Jiaotong University

Digital Signal Processing

Zhou Hang

Comparison of FIR and IIR digital filters



Lancaster University College at Beijing Jiaotong University

- Design principle

IIR digital filter design is realized by designing analog filter;

The analog filters is realized by designing three template low-pass filters;

The low-pass filter is realized by the given boundary frequency and its decay.

FIR digital filter design is realized through time-domain approximation or frequency-domain approximation methods;

Given $H_d(e^{j\Omega})$, determine the type of linear phase DF, $A_d(\Omega)$ and $\varphi_d(\Omega)$;

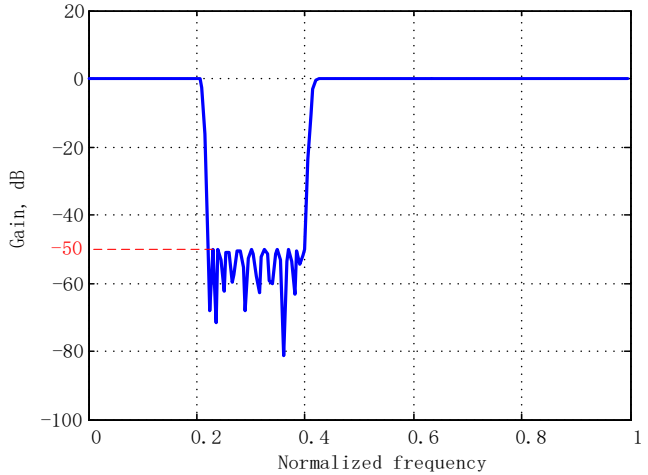
Then the digital filter $H(z)$ can be obtained through the window function method, frequency sampling method, or optimized design.

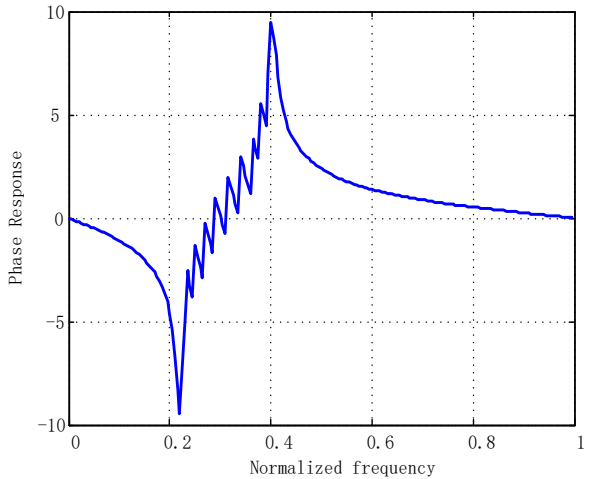
Comparison of FIR and IIR digital filters

➢ Order comparison： Design IIR digital filter and FIR digital filter that meet the following indicators

Ω_{p1}=0.2π rad, Ω_{s1}=0.22π rad, Ω_{s2}=0.4π rad, Ω_{p2}=0.42π rad, $A_p \leq 1$dB, $A_s \geq 50$dB, plot the gain response and phase response, and compare them.

IIR Chebyshev Type II filter Order 11, A_p=0.62dB， A_s=50dB





Comparison of FIR and IIR digital filters

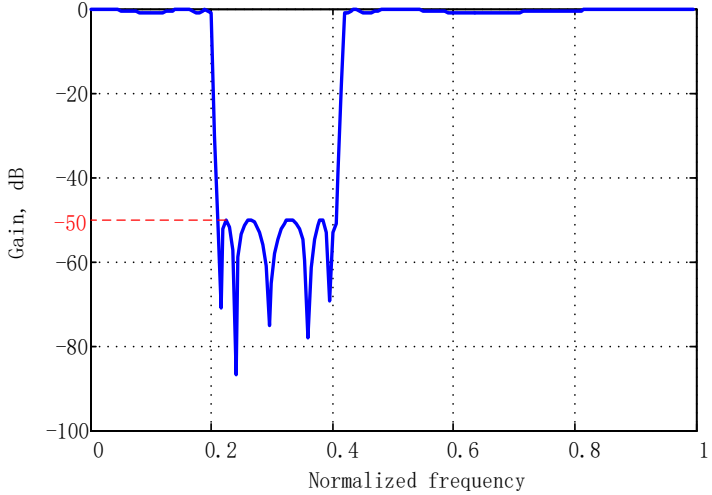
➢ Order comparison： Design IIR digital filter and FIR digital filter that meet the following indicators

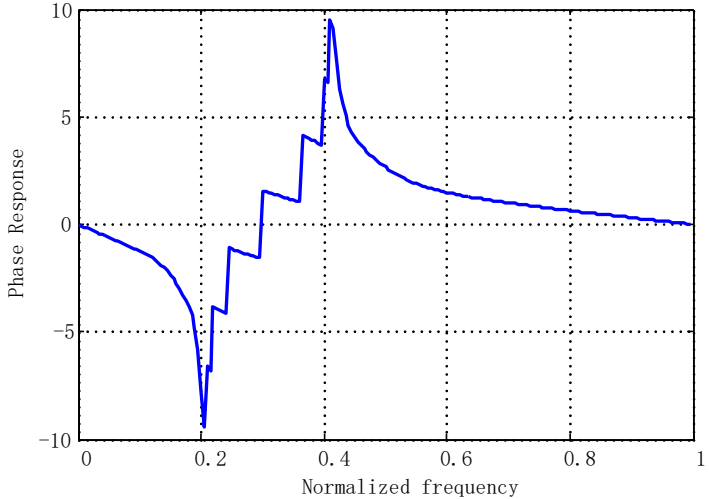
$\Omega_{p1}=0.2\pi$ rad, $\Omega_{s1}=0.22\pi$ rad, $\Omega_{s2}=0.4\pi$ rad, $\Omega_{p2}=0.42\pi$ rad,

$A_p \leq 1$dB, $A_s \geq 50$dB, plot the gain response and phase response, and compare them.

IIR Elliptic filter

Order 6, $A_p=1$dB, $A_s=50$dB





Comparison of FIR and IIR digital filters

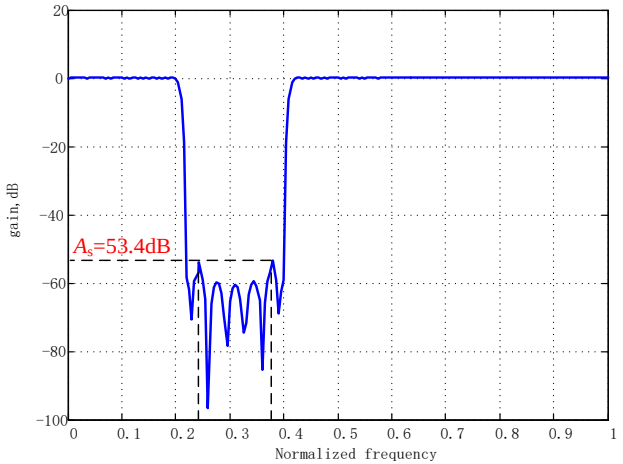
➢ Order comparison： Design IIR digital filter and FIR digital filter that meet the following indicators

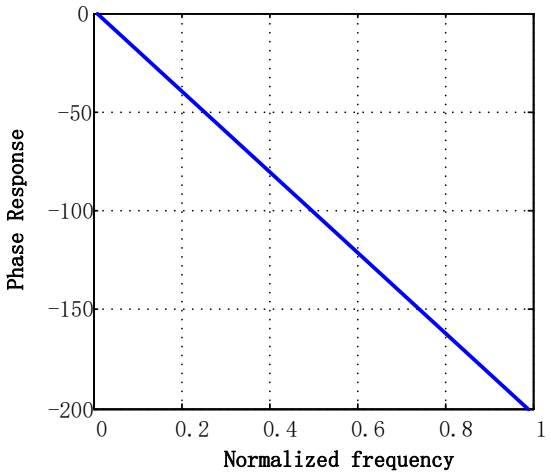
$\Omega_{p1}=0.2\pi$ rad, $\Omega_{s1}=0.22\pi$ rad, $\Omega_{s2}=0.4\pi$ rad, $\Omega_{p2}=0.42\pi$ rad,

$A_p \leq 1$dB, $A_s \geq 50$dB, plot the gain response and phase response, and compare them.

FIR Hamming window

Order 340, $A_p=0.02$dB, $A_s=53.4$dB





Comparison of FIR and IIR digital filters

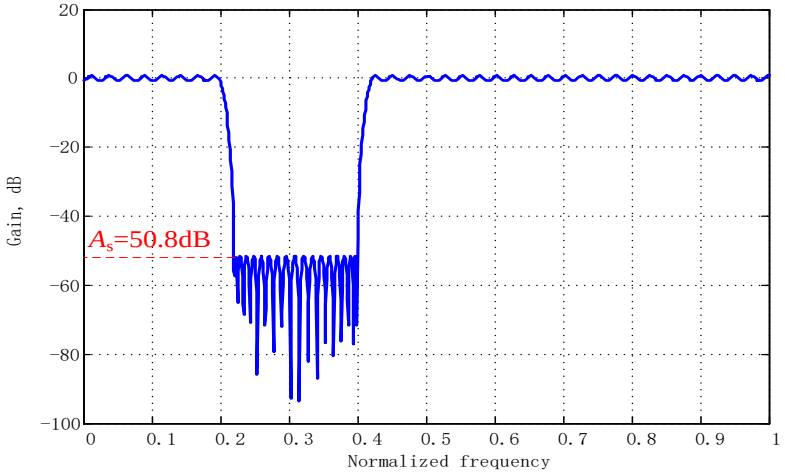
➢ Order comparison： Design IIR digital filter and FIR digital filter that meet the following indicators

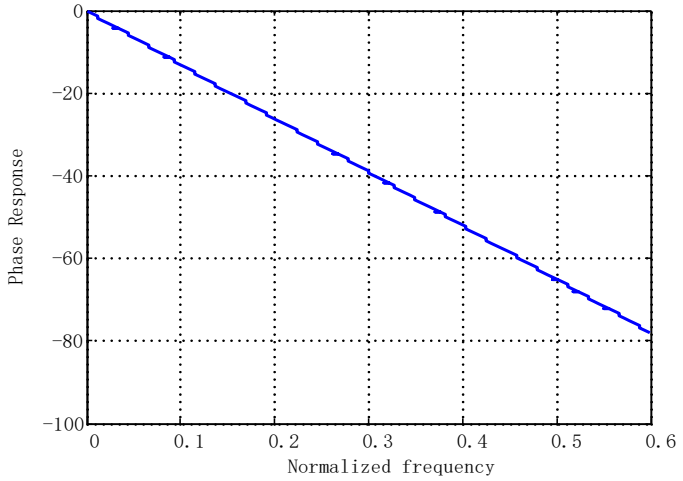
$\Omega_{p1}=0.2\pi$ rad, $\Omega_{s1}=0.22\pi$ rad, $\Omega_{s2}=0.4\pi$ rad, $\Omega_{p2}=0.42\pi$ rad,

$A_p \leq 1$dB, $A_s \geq 50$dB, plot the gain response and phase response, and compare them.

FIR PM algorithm filter

Order 164, $A_p=0.9$dB, $A_s=50.8$dB



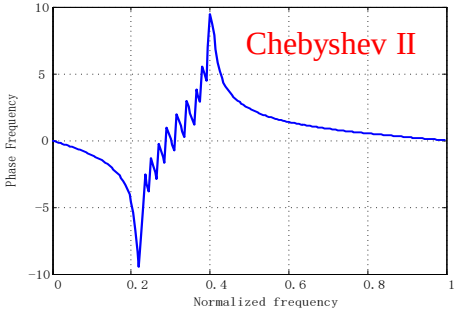


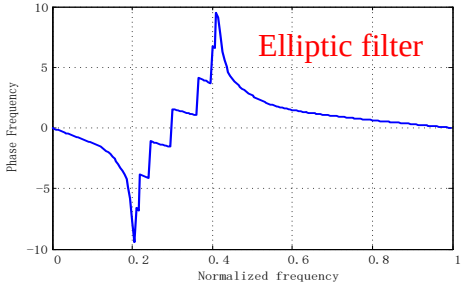
Comparison of FIR and IIR digital filters

Lancaster University College at Beijing Jiaotong University

		Order
IIR	Chebyshev II	**12**
	Elliptic filter	**6**
FIR	Hamming window	**340**
	PM algorithm filter	**164**

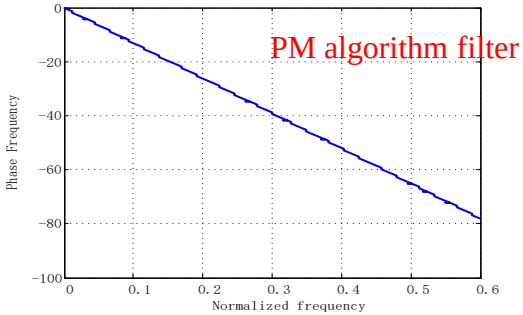
IIR phase response

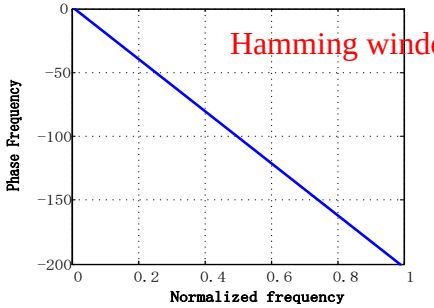




Nonlinear phase

FIR phase response





Linear phase

Comparison of FIR and IIR digital filters

Conclusion:

To achieve the same specifications, the number of orders required for a FIR digital filter is significantly larger than that for an IIR digital filter.

FIR digital filters are easy to implement with linear phase, and the system is also easy to implement.

IIR filters are suitable for processing signals that do not require high phase accuracy, such as speech signals.

FIR filters are suitable for processing signals that require high phase accuracy, such as image signals.

Comparison of FIR and IIR digital filters



Lancaster University College at Beijing Jiaotong University

	IIR DF	FIR DF
stability	uncertain	stable
Implementation difficulty	difficult	easy
Implementing linear phase	no	yes
Can be implemented using FFT	no	yes
Length of $h[k]$	infinite	finite
Order	low	high

Thanks

The materials referenced in this course are accumulated over years of teaching by the instructor, sourced from various media and exchanges with colleagues and peers. It is difficult to indicate the source of each material one by one. This is to clarify and express gratitude.